Biometrics 59, 1123–1130
December 2003

CONSULTANT'S FORUM

Nonidentifiability of Population Size from Capture-Recapture Data with Heterogeneous Detection Probabilities

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Summary. Heterogeneity in detection probabilities has long been recognized as problematic in mark-recapture studies, and numerous models developed to accommodate its effects. Individual heterogeneity is especially problematic, in that reasonable alternative models may predict essentially identical observations from populations of substantially different sizes. Thus even with very large samples, the analyst will not be able to distinguish among reasonable models of heterogeneity, even though these yield quite distinct inferences about population size. The problem is illustrated with models for closed and open populations.

Key words: Capture-recapture; Detection probability; Heterogeneity; Population estimation; Wildlife statistics.

1. Introduction

The problem of making inference about unknown population sizes is of enormous importance in wildlife statistics, with applications in quality control and many other fields. Capture-recapture methods commonly used in wildlife studies consist of combining information about observed animals with plausible models of the observation process, to reach conclusions about the number of animals that have not been observed.

The Lincoln-Petersen method of estimating a population size is one of the oldest and simplest examples: a sample of m_1 individuals from a population is marked and released; later, a second sample is obtained, consisting of u_2 unmarked and m_2 marked individuals. The goal is to estimate N, the population size at the time of the second sample. Treating the ratio of marked to unmarked individuals in the second sample as an estimate of the corresponding population ratio, and assuming that the number of marked animals has not changed between sampling periods, a simple estimator of N is obtained by solving

$$\frac{m_2}{u_2} = \frac{m_1}{N - m_1}.$$

The method can be mathematically formalized, and the precision of the estimated population size assessed, under the further assumption that captures at the second sampling occasion are independent Bernoulli trials with detection probability p for all animals, whether previously marked or not. Here and throughout, it will be convenient to represent the distribution of random variable X by $[X]$, the conditional distribution of random variable X given random variable Y by $[X \mid Y]$, and the binomial distribution with index M and success rate π by $\mathrm{B}(M, \pi)$. With these conventions, the mathematical model for this simple capture-recapture experiment is that m_2 and u_2 are independent random variables, $[m_2] = \mathrm{B}(m_1, p)$ and $[u_2] = \mathrm{B}(N - m_1, p)$. The specification that there are m_1 marked animals at the time of the second sample is usually justified through assumptions that the population is closed (i.e., has not changed because of births, deaths, immigration, or emigration), and that individuals do not lose their marks.

These assumptions can be relaxed, or tested under alternative sampling schemes and using more complex models. A vast literature has developed for estimation of survival rates, recruitment rates, and movement rates using capture-recapture data; of special note is the Jolly-Seber model and its many extensions (see Schwarz and Seber, 1999, for a recent review).

One violation of assumptions that was early identified as especially problematic is the existence of heterogeneity in detection probabilities. The consequences of unmodeled heterogeneity in this nuisance parameter for estimation of other model parameters are not restricted to overstatements of precision, but may also include substantial biases. Burnham and Overton (1978) suggested a jackknife estimator for closed population estimation in the presence of heterogeneous detection rates. Recent wildlife applications have used finite mixture models for the distribution of p (Norris and Pollock, 1996; Pledger, 2000); Coull and Agresti (1999) describe the use of a logit normal model for p; Dorazio and Royle (2003) focus on beta models.

The purpose of this note is to draw attention to a problem inherent to all attempts to model heterogeneity in detection probability. The problem has not gone entirely unnoticed previously, but its extreme consequences appear not to have been fully recognized. The problem is this: that while it may be possible to discern heterogeneity in detection probability, it is likely that an analyst will be unable to distinguish between reasonable descriptions of the heterogeneity, even when these alternative descriptions lead to vastly different inferences about population sizes.

2. Example 1: Closed Population

Suppose that a closed population of size N is sampled on T occasions, that the number of times individual i is observed is X_i, that $X_1, X_2, \ldots, X_N$ are independent random variables with $[X_i \,|\, p_i] = \mathrm{B}(T,\, p_i)$, and that the p_i's are identically distributed according to distribution $g(p)$. The only available data are the frequencies $f_x = \#(X_i = x)$, $x = 1, 2, \ldots, T$, for the $n \leq N$ animals having $X_i > 0$; the problem is to estimate N.

The marginal probability that a randomly selected animal will be sighted x times is

$$\pi_g(x) = \binom{T}{x} \int_0^1 p^x (1-p)^{(T-x)} g(p)\, dp, \quad x = 0, 1, \ldots, T. \quad (1)$$

Letting $f_0 = N - n$, the marginal distribution of vector $\boldsymbol{f} = (f_0,\, f_1,\, f_2 \ldots, f_T)'$ is a $(T+1)$-cell multinomial with index N, and cell probabilities $\boldsymbol{\pi}_g = (\pi_g(0),\, \pi_g(1), \ldots, \pi_g(T))'$; I denote this distribution as $[\boldsymbol{f}\,] = \boldsymbol{M}_{T+1}(N,\, \boldsymbol{\pi}_g)$. It follows that conditional on n, the vector of observed frequencies $\boldsymbol{f^c} = (f_1,\, f_2, \ldots, f_T)'$ has a T-cell multinomial distribution with index n, and cell probabilities $\boldsymbol{\pi_g^C} = \{\pi_g^{\mathrm{C}}(1),\, \pi_g^{\mathrm{C}}(2), \ldots, \pi_g^{\mathrm{C}}(\mathrm{T})\}'$, where

$$\pi_g^{\mathrm{C}}(x) = \frac{\pi_g(x)}{1 - \pi_g(0)}, \quad x = 1, 2, \ldots, T. \quad (2)$$

We may write $[f] = [\boldsymbol{f^c} \,|\, n][n]$, where $[\boldsymbol{f^c} \,|\, n] = M_T(n,\, \pi_{\boldsymbol{g}}^{\mathrm{C}})$ and $[n] = \mathrm{B}\{N,\, 1 - \pi_g(0)\}$. I refer to $\pi_{\boldsymbol{g}}^{\mathrm{C}}$ as the "observable frequency distribution" (OFD), and to $\boldsymbol{\pi_g}$ as the "complete frequency distribution."

Assuming that the distribution g is governed by parameter vector $\boldsymbol{\theta}$, estimation of N can be conducted using the full likelihood $L(\boldsymbol{\theta},\, N; \boldsymbol{f}) \propto [\boldsymbol{f}]$. Alternatively, one might first estimate $\boldsymbol{\theta}$ using the conditional likelihood $L^c(\boldsymbol{\theta}; \boldsymbol{f^c}) \propto [\boldsymbol{f^c} \,|\, n]$, obtain an estimate $\pi_{\hat{g}}(0)$, and define an estimator

$$\hat{N} = \frac{n}{1 - \pi_{\hat{g}}(0)}.$$

The two approaches are asymptotically equivalent (Sanathanan, 1972). The latter highlights the fact that the problem of estimating N is essentially the same as the problem of estimating $\pi_g(0)$, on the basis of information regarding $\pi_{\boldsymbol{g}}^{\mathrm{C}}$. Thus, if g and h are alternative distributions for p, one would hope that $\pi_{\boldsymbol{g}}^{\mathrm{C}}$ close to $\pi_{\boldsymbol{h}}^{\mathrm{C}}$ would imply that $\pi_g(0)$ is close to $\pi_h(0)$. Unfortunately, this is not the case, as I shall demonstrate. The consequence is that inference regarding N is highly model dependent.

2.1 *When Is N an Identifiable Parameter?*

Huggins (2001) describes the parameter N as not identifiable "if different sized populations can give rise to approximately the same [observable] capture histories" and investigates conditions which rule out this possibility. Let G be a set of distribution functions for detection probabilities, and let H be the mapping from G to the set of observable frequency distributions $\pi_{\boldsymbol{g}}^{\mathrm{C}}$. Huggins (2001) states as a necessary condition for identifiability of N that the inverse of H be continuous, so that

$$\boldsymbol{\pi_g^C} \text{ close to } \boldsymbol{\pi_h^C} \Rightarrow g \text{ is close to } h.$$

Assessing "closeness" of distributions requires the choice of a metric; Huggins uses the Hellinger metric. I note two things about Huggins' necessary condition: first, that it is slightly stronger than necessary; what is needed is that

$$\boldsymbol{\pi_g^C} \text{ close to } \boldsymbol{\pi_h^C} \Rightarrow \pi_g(0) \text{ is close to } \pi_h(0).$$

Second, and more important, I note that Huggins' necessary condition includes an implicit assumption that the inverse of mapping H exists. This is in effect an assumption about the set G of distributions considered; it is necessary to limit attention to a set G, and to refer to identifiability of N over G. In particular, N is not identifiable over G if there exist distributions g and $h \in G$ such that

$$\boldsymbol{\pi_g^C} = \boldsymbol{\pi_h^C}, \text{ but } \pi_g(0) \neq \pi_h(0);$$

these two conditions contradict the invertibility of mapping H.

A moment's reflection makes plain that some restrictions on G are necessary: we cannot hope to have N identifiable over the set of all distributions on [0,1]. Given g, one can easily construct a distribution h with $\pi_{\boldsymbol{g}}^{\mathrm{C}} = \pi_{\boldsymbol{h}}^{\mathrm{C}}$ and with any value $\pi_h(0) \in [\pi_g(0), 1]$. This is accomplished by specifying that h is a γ:$(1-\gamma)$ mixture of a point mass on zero and the distribution g, fixing $\pi_h(0)$, and solving $\gamma + (1-\gamma)\pi_g(0) = \pi_h(0)$ for γ. It won't do to dismiss this example on the grounds, sometimes expressed, that "we're only able to make inference for the observable portion of the population" ... move the point mass on zero to mass on a small value p_ε, and the entire population becomes observable, but N is not identifiable.

The foregoing discussion shows that identifiability of N requires restrictions on the class of distributions considered. The example is based on difficulties associated with low-detection probabilities. It is natural to ask whether the problem of identifiability can be resolved by restricting the class of candidate distributions G, so as to rule out the problem of low-detection probabilities. Are low-detection probabilities the only cause of nonidentifiability? I answer this question in the negative in the following example.

2.2 *Low Detection Probabilities Are Not the Only Problem*

Let $0 < \gamma < 1$ be fixed, and let G_γ be the set of probability distributions with support on a subset of $[1 - (1-\gamma)^{1/T},\, 1]$. By restricting attention to $g \in G_\gamma$, we rule out the existence of non-detectable individuals (as in Norris and Pollock, 1996); every individual has probability of at least γ of being detected on one or more sampling occasions.

Let g and h denote two distributions for p, and let g^c and h^c denote the corresponding conditional distributions among

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Table 1

Distinct distributions of p yielding equal conditional marginal distributions. Uniform distribution has parameters $a = 0.026$, $b = 0.80$. *Two-point mixture places masses* {0.576421, 0.423579} *on* {0.286245, 0.676474}.

		0	1	2	3	4
Uniform(a,b)	$\pi_g(x)$	0.227	0.255	0.243	0.190	0.085
	$\pi_g^{\mathrm{C}}(x)$	–	0.329	0.315	0.246	0.110
2 pt. mixture	$\pi_h(x)$	0.154	0.279	0.266	0.208	0.093
	$\pi_h^{\mathrm{C}}(x)$	–	0.329	0.315	0.246	0.110

captured individuals. Huggins (2001), Theorem 2, states that for g, $h \in G_\gamma$,

$$g^c \text{ close to } h^c \Rightarrow g \text{ close to } h.$$

Huggins (2001), Theorem 3, states that

$$g^c \text{ close to } h^c \Rightarrow \boldsymbol{\pi}_{\boldsymbol{g}}^{\mathbf{C}} \text{ close to } \boldsymbol{\pi}_{\boldsymbol{h}}^{\mathbf{C}}.$$

If the converse of his Theorem 3 could be demonstrated (Huggins describes this as "difficult to establish" and "an open question"), then the restriction of attention to G_γ could be taken as sufficient to ensure that the observed frequency distributions characterize the heterogeneity in detection probability: the parameter N would be identifiable over G_γ.

The restriction is not sufficient; in this section, I construct distributions g, $h \in G_\gamma$ with $\pi_{\boldsymbol{g}}^{\mathbf{C}} = \pi_{\boldsymbol{h}}^{\mathbf{C}}$ and $\pi_g(0) \neq \pi_h(0)$: the observable data are identically distributed under the two models, but the inferences about population size are distinct.

Table 1 gives the actual and observable frequency distributions for two models of heterogeneity in detection rates, for the case $T = 4$. Distribution g is uniform on [0.026, 0.800]; distribution h represents a two-point distribution, placing masses {0.576421, 0.423579} on {0.286245, 0.676474}. These distributions are both in G_γ, where $\gamma = 1 - (1 - 0.026)^4 = 0.100$ is the minimum probability of being detected at least once. The observable frequency distributions are identical, even though the complete frequency distributions are substantially different. The two models are indistinguishable based on data for $T = 4$ sampling occasions, yet $\pi_g(0) = 0.227$ and $\pi_h(0) = 0.154$.

The construction of this example may be helpful in illustrating the difficulties associated with relating observed frequency distributions to distributions of detection probabilities. Expanding $(1 - p)^{T-x}$ in equation (1) by the binomial theorem, straightforward calculations yield the relation

$$\pi_g(x) = \sum_{k=x}^{T} \frac{T!}{x!(k-x)!(T-k)!}(-1)^{k-x} m_g(k), \quad x = 0, 1 \ldots, T,$$

where $m_g(k)$ is the kth moment of g. Thus, for $x \geq 1$, the marginal probabilities $\pi_g(x)$ are linear combinations of the first T moments of the distribution of p. Suppose that the moments of g and h are related by $m_h(k) = cm_g(k)$, for $k = 1, 2, \ldots, T$, for some fixed $c \neq 1$. Then for $x \geq 1$, $\pi_h(x) = c\pi_g(x)$, from which it follows that $\{1 - \pi_h(0)\} = c\{1 - \pi_g(0)\}$, and consequently, that the resulting OFDs $\pi_{\boldsymbol{g}}^{\mathbf{C}}$ and $\pi_{\boldsymbol{h}}^{\mathbf{C}}$ are identical.

Thus, to construct the foregoing example, I began by calculating the first $T = 4$ moments of the uniform distribution g on [0.026, 0.800], $m(i)$, $i = 1, 2, 3, 4$. I sought an alternative distribution of detection probabilities, with moments scaled by a factor c. Two-point mixture distributions have 3 parameters, placing masses $\{\psi, 1 - \psi\}$ on $\{p_1, p_2\}$. Equating the first 4 moments of the 2-point mixture with c times the moments of g , one obtains a system of 4 equations in 4 unknowns (ψ, p_1, p_2, and c),

$$\psi p_1^k + (1 - \psi)p_2^k = cm(k), \quad k = 1, 2, 3, 4. \tag{3}$$

The parameters of the 3-point mixture in Table 1 are solutions, to five decimal place accuracy, of equation (3); the value of c was 1.412547.

The point of this example is that the parameter N is not identifiable, even if low-detection probabilities are ruled out. Some further restrictions on the set G of distributions for the detection probabilities must be considered.

This construction uses a mixture distribution governed by a large number of parameters relative to the sample size. As a practical matter, data analysts may wish to rule out models with large numbers of parameters relative to T. Ruling models out of consideration and ruling them out of the realm of possibility are, of course, different matters. Supposing that a data set with $T = 4$ and a large value of n were analyzed assuming that p's were uniformly distributed on$[a, b]$, and the fit found to be extraordinarily good for $\{\hat{a} = 0.026, \hat{b} = 0.80\}$, there would be no way of knowing which of the two models in Table 1 were correct, even though there is a 20% discrepancy in the proportions used to scale n to an estimate of population size.

Still, it is natural to ask whether the difficulty identified here is merely a pathology associated with highly parameterized models. The next examples show that distinct models governed by only a few parameters can produce practically identical OFDs $\boldsymbol{\pi}_{\boldsymbol{g}}^{\mathbf{C}} \approx \boldsymbol{\pi}_{\boldsymbol{h}}^{\mathbf{C}}$, while producing substantially different inferences for N, because $\pi_g(0) \neq \pi_h(0)$. Restricting attention to models described by relatively fewer parameters does not solve the identifiability problem.

2.3 *Restricting Attention to Models Governed by Fewer Parameters is Not the Answer*

Suppose that $T = 5$, and that the logit(p_i)'s are normally distributed with mean $\mu = -2$ and standard deviation $\sigma = 1$. The complete and observable frequency distributions, $\boldsymbol{\pi}_{\mathbf{1}}$ and $\boldsymbol{\pi}_{\mathbf{1}}^{\mathbf{C}}$, are given in Table 2 (calculated by simulation to 3-decimal-place accuracy). In particular, note that $\pi_1(0) = 0.507$; roughly half of the population will be observed at least once.

An analyst would likely begin by fitting a model without heterogeneity in detection rates, that is, with constant $p = p_0$. We may calculate a best approximating model under this, or any other, parametric misspecification g_M of the distribution of p, by supposing that data $f_x = n\pi_1^{\mathrm{C}}(x)$, $x = 1, 2, \ldots, T$ correspond exactly to the expected cell frequencies of the true model, then maximizing the expected log likelihood as a function of the parameters governing g_M. This is equivalent to

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Table 2
Marginal distributions $\pi_i(x) = \Pr(x = x)$ *and conditional marginal distributions* $\pi_i^{\mathrm{C}}(x) = \Pr(X = x|X > 0)$ *for four different models of heterogeneity in mixed binomial data.*

		0	1	2	3	4	5	λ_M/n
1) Logit-Normal $\mu = -2$, $\sigma = 1$	$\pi_1(x)$	0.507	0.296	0.129	0.049	0.015	0.003	–
	$\pi_1^{\mathrm{C}}(x)$	–	0.602	0.262	0.099	0.301	0.006	
2) Homogeneous $p_0 = 0.230$	$\pi_2(x)$	0.270	0.404	0.242	0.072	0.011	0.001	0.046
	$\pi_2^{\mathrm{C}}(x)$	–	0.554	0.331	0.099	0.015	0.001	
3) Beta $a = 0.751$, $b = 5.105$	$\pi_3(x)$	0.594	0.245	0.106	0.041	0.013	0.002	6.6e-5
	$\pi_3^{\mathrm{C}}(x)$	–	0.602	0.260	0.101	0.031	0.006	
4) 2-point mixture $p_1 = 0.147$,	$\pi_4(x)$	0.399	0.361	0.158	0.059	0.019	0.003	1.8e-5
$p_2 = 0.480$, $\psi = 0.871$	$\pi_4^{\mathrm{C}}(x)$	–	0.601	0.263	0.097	0.032	0.006	

λ_M is the Kullback-Leibler discrepancy of $\pi_i^{\mathrm{C}}(x)$ from $\pi_1^{\mathrm{C}}(x)$.

minimizing $2n$ times the Kullback-Leibler (KL) distance from $\boldsymbol{\pi}_1^{\mathbf{C}}$ to π_M^{C},

$$\lambda_M = 2n \sum_{x=1}^{T} \pi_1^{\mathrm{C}}(x) \ln \left\{ \frac{\pi_1^{\mathrm{C}}(x)}{\pi_M^{\mathrm{C}}(x)} \right\}; \tag{4}$$

λ_M is the noncentrality parameter of the likelihood ratio goodness of fit statistic (Agresti, 1990, p. 241).

For the logit-normal model already described, the best approximating homogeneity model is found to have $p_0 = 0.230$; the corresponding OFD is given in Table 2 as $\pi_2^{\mathrm{C}}(x)$; the noncentrality is 0.046 n. Using tables of the non central chi-squared distribution, we find that for $n = 130$, the power of an $\alpha = 0.05$ test will only be about 1/2, and that for $n < 200$, the power is less than 0.67. An analyst that does not want to be concerned with heterogeneity and has n less than 200, will have a fairly good chance of concluding that the homogeneous model is acceptable, and to conclude that all but roughly a quarter of the population has been observed ($\pi_2[0] = 0.27$), leading to a negative bias of 32.5% in estimation of N. I wholly concur with a referee's comment that "it is a very poor analyst who will use the lack of power from a small data set to conclude that the homogeneous model is acceptable ... [because] there is likely to be underestimation of N." Whether detected or not, how will the analyst choose to model the heterogeneity?

The beta distribution is appealing, since the integral (1) can be evaluated in terms of the gamma function. Parameter estimation cannot be conducted in closed form, but numerical minimization of λ_M over the beta family of densities is straightforward. The best fitting OFD is tabulated in Table 2 as $\pi_3^{\mathrm{C}}(x)$. The fit is not perfect, but close enough so that λ_M is roughly $6.6n/100{,}000$; there is virtually no power to distinguish the beta and logit-normal models, except with very large samples.

Alternatively, an analyst might suppose that the distribution g in (1) is a 2-point mixture distribution, one of a class of mixture models suggested by Norris and Pollock (1996). This model posits that the detection probability either is p_1 with probability ψ, or is p_2, with probability $(1 - \psi)$. The model is appealing, not because a two (or 3, or 4, ...) point mixture is likely to be exact, but because there is a step-function, nonparametric flavor to it; the mixture might be thought of as approximating a continuous g. The best approximating 2-point mixture model is given in Table 1, as $\pi_4^{\mathrm{C}}(x)$. Once again, unless n is enormous, there will be no power to distinguish the fitted model from the logit-normal.

The results of Table 1 show that while it may be possible to obtain evidence that heterogeneity exists, it may be impossible to choose among plausible models of its form. This would be a matter of little consequence, except that the fitted values of $\pi(0)$, and consequently inferences about N, are quite inconsistent among models.

A referee suggested that the discrepancies among values of $\pi(0)$ might not matter so much, because for moderate sample sizes, interval estimates obtained under any of the models considered are long and have similar lower limits. Indeed, bias is not a very great concern, if it is small relative to mean squared error. However, the acceptability of biased estimators usually requires some assumption that the bias disappears at a rate proportional to that of the variance as sample size increases. If it does not, confidence interval coverage will approach zero as sample size increases. For the models described in Table 1, the bias does not disappear as sample size increases; for large enough n, contradictory inferences are inevitable. The only question is whether with such large n, we will be perhaps be able to reject the fit of one model or the other.

My final example is a data set that fits several distinct models remarkably well, models that produce contradictory inferences, with nonoverlapping confidence intervals. The example shows that these problems persist, even with large values of T and n, and that data insufficient to distinguish between models can lead to contradictory inferences about N.

2.4 *Problems Remain for Larger Values of T*

Consider the data set: $\boldsymbol{f} = (679, 531, 379, 272, 198, 143, 99, 67, 46, 32, 22, 14, 9, 5, 3, 1, 0, 0, 0, 0)'$, with $n = 2500$ and $T = 20$. Table 3 reports analyses of the data under 7 distinct models for detection probabilities A: homogeneous detection rates; B, C, and D: 2, 3, and 4-point mixture models; E: a logit normal model, F: a log-gamma model, and G: a beta model.

While the homogeneous model and a 2-point mixture model are clearly unacceptable in terms of their fit, the remaining 5 all fit well. Models E, F, and G fit extraordinarily well, are described by only two parameters, and are favored by Akaike's information criterion, AIC = −2(max log likelihood − #parameters). These models nevertheless provide contradictory inferences about N; the 95% profile likelihood intervals obtained under models E and G do not overlap. Thus, while

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Table 3
Six analyses of data $\boldsymbol{f} = (679, 531, 379, 272, 198, 143, 99, 67, 46, 32, 22, 14, 9, 5, 3, 1, 0, 0, 0, 0)'$, *with* $n = 2500$, $T = 20$.

	$\hat{N}$	95% PLI	Δ AIC	χ^2	*df*	*p*-value
A	2571	(2554, 2589)	1661.4	2.5e5	13	<0.0001
B	2776	(2730, 2827)	121.04	205.6	11	<0.0001
C	2930	(2840, 3025)	7.56	4.36	9	0.8762
D	2992	(2867, >10000)	7.42	0.21	7	>0.9999
E	3111	(3018, 3218)	1.72	2.40	12	0.9985
F	3111	(3174, 3477)	0	0.94	12	>0.9999
G	3494	(3308, 3730)	1.28	2.71	12	0.9973

A: Homogeneous detection probability, $p = 0.164$
B: 2-point mixture, placing masses {0.775, 0.225} on {0.098, 0.339}
C: 3-point mixture, placing masses {0.623, 0.322, 0.055} on {0.071, 0.229, 0.480}
D: 4-point mixture, placing masses {0.523, 0.328, 0.123, 0.026} on {0.058, 0.172, 0.326, 0.538}
E: logit(p) $\sim$ Normal, mean = -2.172, standard deviation = 0.986
F: log($1/p$) $\sim$ Gamma, mean = 2.514, standard deviation = 1.087
G: $p \sim$ Beta ($a = 0.888$, $b = 6.458$)
$\hat{N}$ is the maximum likelihood estimator of N, 95% PLI is a 95% profile likelihood interval, ΔAIC is the difference between the model's maximum likelihood and that of model F. χ^2, *df*, and *p*-value summarize a Pearson's chi-squared goodness-of-fit test. Pearson chi-squared values computed after pooling cells 14-20.

the parameter N might be identifiable, subject to the restriction to two parameter models, it is for all practical purposes not identifiable, even with this restriction.

The sample size in the example of Table 3 being fairly large, I conducted a simulation study to evaluate the effect of model misspecification for smaller values of n. I used a logit normal model with $\mu = -2.185$ and $\sigma = 1$ for the detection probabilities. For each sample size $n = 250$, 500, 750, 1000, 1500, and 2000, I generated 1000 data sets, and estimated confidence interval coverage rates for estimates of $1/\{1 - \pi(0)\}$ obtained under the beta model. As might be anticipated from the results of Table 3, the beta model estimates were typically too large (exceeding the true value 82% of the time for $n = 250$, and nearly 94% of the time for $n = 2000$). Nominal 95% confidence intervals had coverage of 98% for $n = 250$, but coverage rates dropped to 71%, 55%, 46%, 32%, and 26% as n increased to 500, 750, 1000, 1500, and 2000.

The results in Table 3 suggest that, given heterogeneity exists, analyses allowing for it are likely to produce better estimates of N than analyses that do not, even if the form of the heterogeneity is misspecified. Nevertheless, undetectable model misspecification leads to biased estimation and less-than-nominal confidence interval coverage.

2.5 *Summary of Example 1*

The examples of Section 2.2 show that even if we rule out the existence of non observable individuals, the parameter N is not identifiable. The examples of Sections 2.3 and 2.4 show that even if identifiability of N could be attained by restricting attention to heterogeneity models governed by two or three parameters, for practical purposes, the problem remains; it is very difficult to distinguish between models yielding substantially different inferences, even with large values of T. The simulation study in Section 2.4 illustrates that model misspecification leads to biased estimation and less-than-nominal confidence interval coverage.

Taken together, these examples confirm an empirical observation of Cormack (1979) that "many different forms can be found to fit the truncated distribution, while giving rise to vastly different estimates for the zero-frequency class." Indeed, the parameter N is not identifiable, without recourse to untestable model assumptions.

3. Example 2: Open Population

The Cormack-Jolly-Seber (CJS) model produces estimates of survival rates for open populations based on the frequencies of distinct recapture histories in cohorts of marked individuals. The CJS model posits that all marked individuals have the same probability of capture, p_t, at sampling occasion t. Jolly and Seber extended the model by supposing that the capture probabilities apply to all individuals, marked or unmarked. Then, the number of captures n_t is related to the population size N_t by $\mathrm{E}(n_t) = N_t p_t$. Given an estimate $\hat{p}_t$, the population size is estimated by $\hat{N}_t = n_t/\hat{p}_t$.

I now consider a similar extension of the CJS model, with individual heterogeneity in detection probabilities; my goal is to illustrate the existence of problems similar to those documented for the closed-population analyses. For ease of presentation, my description of the problem and methods is based on a single cohort of n individuals captured, marked, and released at time $t = 0$. I will suppose that the probability of surviving from one sampling occasion to the next is a constant, φ, varying neither among individuals nor through time, and that detection probabilities are constant through time, but vary among individuals. The heterogeneity in detection probability will be modeled by supposing that the individual values p_i, $= 1, 2, \ldots, N$ are sampled from distribution $g(p)$; this distribution describes the entire population of size N at time $t = 0$, and not simply the individuals captured.

The population size N can be estimated in a manner similar to that proposed by Jolly and Seber, on the basis of the relation $\mathrm{E}(n) = N\mathrm{E}(p)$. Care is required to model the distribution

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of detection rates, since under heterogeneity, the distribution of p given capture at time $t = 0$, $g_c(p)$, cannot be the same as the population distribution $g(p)$. Letting ω_{0i}, denote the event that individual i is captured at time $t = 0$, the relation between these distributions is established by noting that

$$g_c(p)dp = \Pr(p_i \in [p, p + dp) \,|\, \omega_{0i} = 1)$$
$$= \frac{p}{\mathrm{E}(p_i)}\Pr\{p_i \in [p, p + dp)\} = \frac{p}{\mathrm{E}(p_i)}g(p)\, dp,$$

so that

$$\mathrm{E}_c\left(\frac{1}{p}\right) = \frac{1}{\mathrm{E}(p)}, \tag{5}$$

where the expectations are with respect to $g_c(p)$ and $g(p)$, respectively. Thus, the population of size N can be estimated by specifying a parametric form for $g_c(p)$, estimating its parameters, calculating the corresponding estimate of $\mathrm{E}_c(1/p)$, and defining

$$\hat{N} = n\hat{\mathrm{E}}_c(1/p).$$

The future history of an individual captured at $t = 0$ is a multinomial random variable with index 1: its history will be one of the 2^T distinct vectors $\boldsymbol{\omega} = (\omega_1, \omega_2, \ldots, \omega_T)'$ where $\omega_t \in \{0, 1\}$, for $t = 1, 2, \ldots, T$. Under the assumptions stated above, the probability of history $\boldsymbol{\omega}$ for individual i, is

$$\pi_\omega(p_i) = p_i^k(1 - p_i)^{T-f-k}\varphi^{T-f}\chi_{f,i},$$

where $k = k(\boldsymbol{\omega}) = \#(1\text{'s in history } \boldsymbol{\omega})$, $f = f(\boldsymbol{\omega}) = \#(\text{final } 0\text{'s in history } \boldsymbol{\omega})$; the parameters $\chi_{f,i}$ are defined recursively as

$$\chi_{f,i} = (1 - \varphi) + \varphi(1 - p_i)\,\chi_{f-1,i},$$

with $\chi_{0,i} \equiv 1$. Thus, the conditional frequency distribution for the 2^T distinct vectors $\boldsymbol{\omega} = (\omega_1, \omega_2 \ldots, \omega_T)'$ among n individuals captured at time $t = 0$ is multinomial with cell probabilities

$$\pi_\omega = \int_0^1 \pi_\omega(p_i)g_c(p_i)\, dp_i.$$

Relation (5) shows that the choice of a parametric form for $g_c(p)$ is constrained by the requirement that $\mathrm{E}_c(1/p)$ must be finite. The choice must also be constrained by its implications for the population distribution $g(p)$. These constraints are nicely dealt with by supposing that $g(p)$ is a member of a parametric family conjugate to a successful Bernoulli trial. Thus, for instance, if the population distribution of p is $\beta(a, b)$, the distribution of p to be modeled among the captured animals is also in the beta family, but with parameters $a_c = a + 1$, and $b_c = b$, so that

$$\mathrm{E}_c\left(\frac{1}{p}\right) = \frac{1}{\mathrm{E}(p)} = \frac{a + b}{a} = \frac{a_c + b_c - 1}{a_c - 1}.$$

Similarly, if the population distribution of p puts point mass of ψ_k on p_k, $k = 1, 2, \ldots, K$, the distribution for captured animals is concentrated on the same values p_k, but with masses ${\psi^c}_k$,

$$\psi_k^c = \frac{\psi_k p_k}{\sum_j \psi_j p_j}.$$

It follows that

$$\psi_k = \frac{\left(\psi_k^c/p_k\right)}{\sum_j \left(\psi_j^c/p_j\right)},$$

so that

$$\mathrm{E}_c\left(\frac{1}{p}\right) = \frac{1}{\mathrm{E}(p)} = \frac{1}{\sum_k \psi_k p_k} = \sum_j \left(\psi_j^c/p_j\right).$$

Another family of distributions exhibiting conjugacy to a successful Bernoulli trial is defined by supposing that the population distribution of $-\log(p)$ is in the gamma family. Then,

$$g(p) = \frac{\lambda^\beta}{\Gamma(\beta)}\{-\log(p)\}^{\beta-1}p^{\lambda-1}, \beta, \lambda > 0, \quad 0 < p < 1,$$

and the relation between parameters of $g(p)$ and $g_c(p)$ is that $\beta_c = \beta$, and $\lambda_c = \lambda + 1$. Thus,

$$\mathrm{E}_c\left(\frac{1}{p}\right) = \frac{1}{\mathrm{E}(p)} = \left(\frac{\lambda + 1}{\lambda}\right)^\beta \left(\frac{\lambda_c}{\lambda_c - 1}\right)^\beta.$$

In light of Example 1, it is relevant to ask how sensitive the estimation of N is to modeling assumptions regarding the distribution of p. Table 4 presents multinomial cell frequencies under logit normal, log gamma, beta, two-point mixture, and homogeneous models of detection probabilities for a study based on $T = 4$ recapture occasions. Model parameters were chosen to minimize the discrepancy

$$\lambda_M = 2n\sum_{j=1}^{t} \pi(\omega) \ln\left\{\frac{\pi(\omega)}{\pi_M(\omega)}\right\}$$

between each model and the logit normal model. For the problem presented here, values $\lambda_M =$ 1, 10, and 25 correspond roughly to power of 0.08, 0.50, and 0.95 for the likelihood ratio goodness of fit test with significance level $\alpha = 0.05$. A sample size of several hundred would be required to reject the homogeneous model. However, $n = 86{,}000$ would only suffice to yield power of 0.50 for distinguishing the logit normal and beta models.

The four heterogeneity models are virtually indistinguishable. Given that any of the four heterogeneity models is the truth, the other three will produce acceptable estimates of survival but use of the homogeneity model would results in a substantial negative bias. However, the factors $\mathrm{E}_c\ (1/p)$ vary widely among the heterogeneity models; if the logit normal were truth, biases in estimation of N are 28.7%, 15.7%, and −20.0%, for the beta, log gamma, and two-point distributions, respectively. Expressed as a proportion of $N - n$, these biases are 35.1%, 19.2%, and −24.5%.

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Table 4
Cell probabilities π_ω for single-cohort, four sampling occasion capture-recapture data as described in example 2, under alternative models for $g_c(p)$.

History (ω)	Logit normal	Beta	Log gamma	Two-point	Homogeneous
0000	0.4467	0.4480	0.4476	0.4453	0.4070
1000	0.1025	0.1013	0.1016	0.1039	0.1389
0100	0.0752	0.0743	0.0746	0.0761	0.0847
0010	0.0608	0.0600	0.0603	0.0615	0.0568
0001	0.0524	0.0517	0.0519	0.0529	0.0424
1100	0.0380	0.0382	0.0381	0.0374	0.0508
1010	0.0278	0.0282	0.0281	0.0271	0.0340
1001	0.0233	0.0236	0.0235	0.0224	0.0254
0110	0.0278	0.0282	0.0281	0.0271	0.0340
0101	0.0233	0.0236	0.0235	0.0224	0.0254
0011	0.0233	0.0236	0.0235	0.0224	0.0254
1110	0.0224	0.0225	0.0225	0.0234	0.0204
1101	0.0178	0.0181	0.0180	0.0188	0.0152
1011	0.0178	0.0181	0.0180	0.0188	0.0152
0111	0.0178	0.0181	0.0180	0.0188	0.0152
1111	0.0232	0.0224	0.0227	0.0216	0.0091
$(10^5/n)\lambda_M$	–	11.6	5.13	51.4	4095.7
$\mathrm{E}_c(1/p)$	5.49	7.09	6.35	4.39	2.67

Parameters for the logit normal model: μ −1, $\sigma = 1$, $\varphi = 0.90$; for the beta model: $a_c = 1.6138$, $b_c = 3.7228$, $\varphi = 0.9009$; for the log gamma model: $\lambda_c = 2.4028$, $\beta_c = 3.4356$, $\varphi = 0.9007$; for the two-point model: $p_1 = 0.1744$, $p_2 = 0.5563$, $\psi_c = 0.6597$, $\varphi = 0.8984$; for the homogeneous model: $p_0 = 0.3747$ and $\varphi = 0.8247$. Asymptotic noncentralities λ_M measure discrepancies of models from the logit normal; $\mathrm{E}_c(1/p)$ is the scaling factor for estimating N from n.

The close agreement of the fitted densities is illustrated in Figure 1.

4. Discussion

The examples presented here illustrate that there are extreme difficulties associated with estimating population size, whether in closed or open-population models, in the presence of heterogeneous detection probabilities. It may be impossible to distinguish between alternative models of the heterogeneity, even with very large data sets, despite the fact that the different models produce substantially different inferences about population sizes.

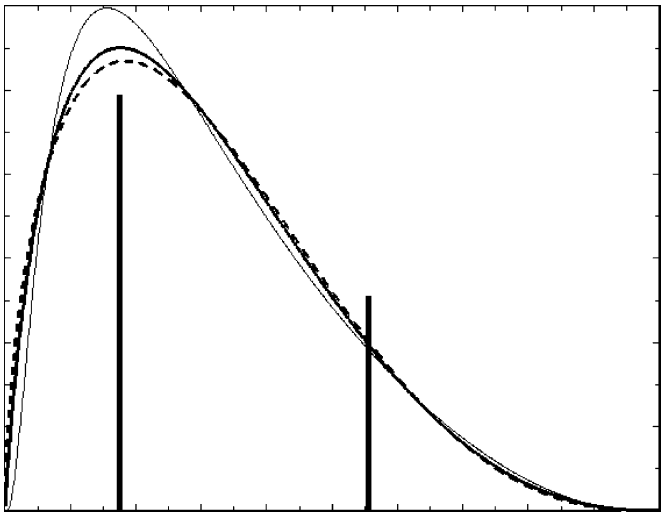


Figure 1. Logit normal (narrow line), beta (thick, solid line), log gamma (thick, dashed line), and point mass distribution (bars, approximately scaled to the continuous densities) corresponding to Example 2.

Analysts should be aware of this phenomenon; at the very least, the effect of choosing alternative parametric models for heterogeneity in detection should be considered. The examples strongly call into question the usefulness of nonparametric methods; they simply cannot be reliable in all cases.

Even given samples of the actual detection probabilities, it would be difficult to distinguish among the three continuous densities in Figure 1. The primary differences among them are in the amount of mass near zero, which plays an important role in the estimation of the number of animals not detected. This observation suggests that similar difficulties are to be anticipated in the use of distance sampling models (Buckland et al., 2001), in which inference is based on a density value at zero.

Acknowledgements

I thank Shirley Pledger and an anonymous referee for their careful reviews. I am grateful for helpful conversations and encouragement from James D. Nichols, Robert M. Dorazio, and J. Andrew Royle.

Résumé

L'hétérogénéité des probabilités de capture constitue une difficulté connue des études de capture-recapture, et de nombreux modèles ont été développés pour permettre d'en tenir compte. Nous montrons par des simulations de données sous différents scénarios que l'hétérogénéité individuelle est

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particulièrement problématique. En effet, sous divers modèles raisonnables d'hétérogénéité individuelle, les données générées peuvent finalement être très semblables, alors même que les effectifs de population sont très différents. Ainsi, même avec de très grands échantillons, l'analyse des données de capture-recapture ne permet pas de distinguer entre différents modèles d'hétérogénéité, même si les effectifs des populations concernées sont très différents. Nous illustrons ce problème avec des modèles pour populations ouvertes ou fermées.

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Received February 2002. *Revised May* 2003.
Accepted June 2003.

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